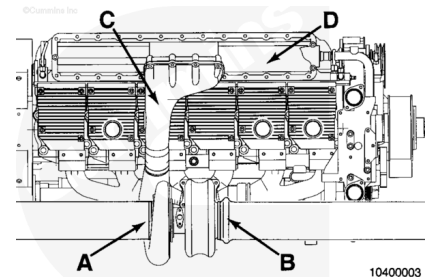


Trainings ▾

General Information

The combustion air system consists of the following:

- A. Turbocharger (intake)
- B. Turbocharger (exhaust)
- C. Air crossover piping
- D. Aftercooler assembly.

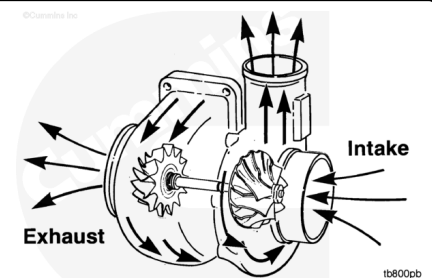


The aftercooler assembly (or intake manifold) to cylinder head mounting capscrews are covered by rubber sleeves. The rubber sleeves are located on the top mounting capscrews at the cylinder head. These sleeves prevent capscrew damage by keeping dirt and water off of the threads. These sleeves **must** be replaced during rebuild.

All QSK19 engines have turbochargers. The turbochargers also have water cooled bearing housings. This gives the turbocharger increased working life by reducing the harm of **hot** shutdowns.

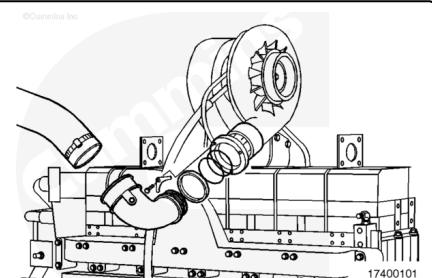
The turbocharger uses exhaust gas energy to turn the turbine wheel. This turbine wheel drives a compressor impeller that provides pressurized air to the engine for combustion.

QSK19 engines can have either jacket water after cooled air cooling systems, or low temperature aftercooling (LTA) air cooling systems. These systems use engine coolant to cool the inlet air between the turbocharger and the engine.



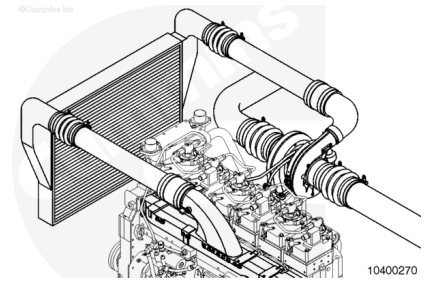
Rail Applications

Because the QSK19 Rail engine uses an air-to-air aftercooler, there is no aftercooler element in the intake manifold that uses coolant to cool the intake air. The intake housing is a one-piece component that contains an air inlet duct. This air-to-air aftercooler has a turbocharger outlet elbow which is unique to all QSK19 air-to-air cooled engines.



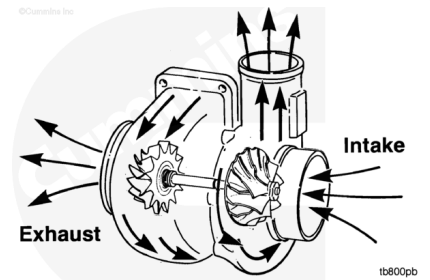
The combustion air system consist of the following:

- Intake air piping
- Turbocharger
- Charge air cooler piping
- Exhaust piping.



⚠ CAUTION ⚠

The correct turbocharger must be used. Providing too much additional air will increase the cylinder pressures and shorten the life of the engine.



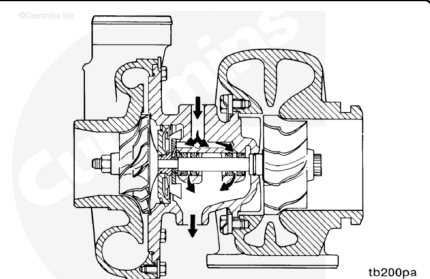
The turbocharger uses exhaust gas energy to turn the turbine wheel. The turbine wheel drives the compressor impeller that provides pressurized air to the engine for combustion. The additional air provided by the turbocharger allows more fuel to be injected to increase the power output from the engine.

Note : An adequate supply of good filtered oil is very important to the life of the turbocharger.

The turbine and compressor wheels are attached to a common shaft and are supported by two rotating bearings in the bearing housing. Passages in the bearing housing direct filtered pressurized engine oil to the shaft bearings and thrust bearings. The oil is used to lubricate and cool the rotating components to provide for smooth operation. The oil then drains from the bearing housing to the engine sump through the oil drain line.

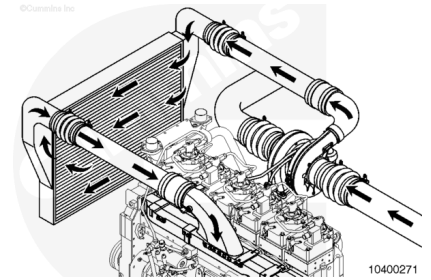
Additional passages in the bearing housing direct engine coolant around the turbine shaft end and shaft bearing. This is to aid in keeping the turbocharger cool and prevent overheating of the turbine wheel and shaft bearing.

A speed sensor located in the bearing housing allows the ECM to monitor turbocharger speed and prevent overspeeding by derating the engine.



As the intake air is compressed by the turbocharger, the air temperature increases. This heated air is then passed through the charge air cooler, which cools the air. Cool air is more

dense, which allows more air to be compressed into the cylinder, yielding a much greater combustion efficiency.



with Electronically Actuated Injector

These engines are equipped with a water cooled turbocharger turbine housing as well as the water cooled bearing housing. This gives the turbocharger increased working life by reducing the harm of hot shutdowns.

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